

POST TENSIONING FORMAT FOR PSC GIRDER

General information:

Name & Location of Bridge: Construction Of 301.00m Long PSC Girder Bridge Over the River Kartoa at Ch 4500 on Chatra GC-Gilabari Ghat Via Nischintobati Primary School Road, Road ID : 185764034		District: Rangpur	Upazila: Pirganj
Span/Girder Ref: Span-1, Girder-A	Girder Length (m): 43m	Cable Ref. to be Stressed: 01 of 04	
Girder Casting Date: 12-07-2026	Stressing Stage of Cable : 02	Jacking end Ref: End 1: L709047	End 2: L709046
Date of Tensioning: 06-08-2026			

Design information: 06-08-2026

Dia of Strand (mm): 15.24mm	No. of Strand Cable: 19 nos	Anchorage Brand: Decorate	UTS of Strand (N/mm²): 1860	M. of Elasticity, E (MPa): 197000
Area of Strand (mm²): 140	Area of Cable, A (mm²): 2660	Design Jacking Force P: 3750 Kn		
Design Elongation Each end (mm):	Design Elongation for Gripping Length (mm): N/A	Actual Gripping Length (mm):	Corrected Elongation for Grip Length d (mm):	
Design Cable Slip (mm): 6mm	Design Conc. St. during Tensioning (N/mm²): 36 MPa	Actual Conc. St at the time of Tensioning (N/mm²): 38.06 MPa		

Stressing & Jack information:

Pump Model No. P1: P2:	Pressure gauge Model: M1: M2:	Tensioning Ram Area (m²): TA1: TA2:	Blocking Ram Area (Cm²): BA1: BA2:
Actual Area of Cable, A1 (mm²): 2707.50	Modulus of Elasticity, E1(MPa): 200400	Corrected Elongation (mm) for Actual A1 & E1: $\delta \cdot (A \cdot E / A1 \cdot E1)$: 141.32 mm	
Jack Pres. With Jack loss (BAR) kg/cm²: J1: P/(TA1*efficiency) J2: P/(TA2*efficiency)	Calibrated Jack press. (kg/cm²): CJ1: CJ2:	Initial Jack Press. (kg/cm²): ICJ1: 87.22 ICJ2: 85.87	Initial Marking: IRJ1: 40 IRJ2: 38
Actual calculated Elongation for Grip Length, d1 (mm): 141 mm	Gross Slip of Cable (mm): (Final Elong - Net Elong)	Net Slip at Jack end after lock-off (mm): (Gross Slip - Calcul. Elong. d1 for Grip length)	
Blocking Pressure kg/cm²: E1: Auto Block E2: Auto Block	At end 1	At end 2	At end 1
	159-152=7	154-146=8	7-4.14=2.86
			8-4.14=3.86

Record of Stressing & Elongation:

Record of Stressing & Elongation:															
Avg. % of Design Load	Actual Applied Pressure		Calculated Gauge Pressure Kg/cm2		Reading for Elongation (mm)		Measured Elongation At both Jacking end (mm)			Correction Factor For ICJ	Final/ Total Elongation. (mm)	Average Elongation At each End.	Remarks (Average Slip at each end)		
	Col(1)		Col(2)		Col(3)		Col(4)=(Col 3-IRJ)							Col(5)	Col(6)=(4+5)
	1	2			1	2	1	2	Avg.					1	2
	KN	TON	Kg/cm2	Kg/cm2	-	-	-	-	-		-	-			
20	750	76.48	87.22	85.87	40	38	-	-	-				2.86+3.86		
40	1500	152.96	175.43	175.53	75	68	35	30	32.5				2		
60	2250	229.44	263.65	265.19	106	103	66	65	65.5				= 3.36		
80	3000	305.91	351.86	354.85	134	127	94	89	91.5						
95	3562.5	363.27	418.01	422.09	159	154	119	116	117.5	30		147.5	< 6		
98	3675	374.75	431.25	435.09											
100	3750	382.39	440.07	444.50									OK		
102	3825	390.04	448.89	453.43											
105	3937.5	401.51	462.12	466.92											
Lock-off					152	146									

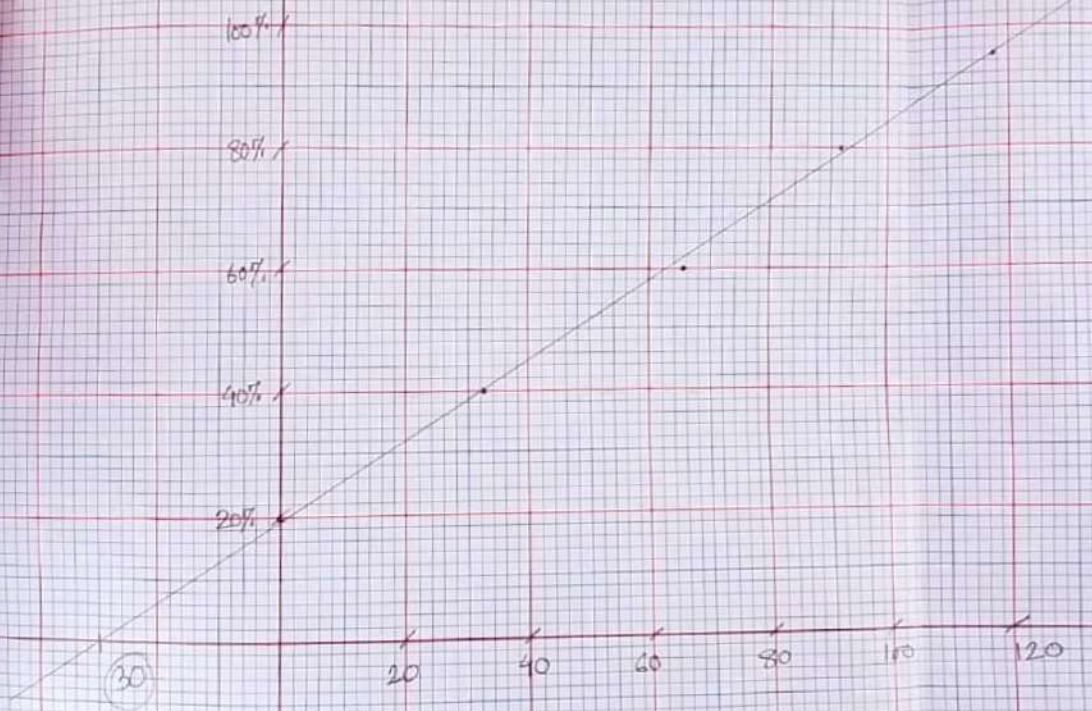
Client's Representative

Consultant's Representative

Contractor's Representative



01 of 04



POST TENSIONING FORMAT FOR PSC GIRDER

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Name & Location of Bridge: Construction Of 301.00m Long PSC Girder Bridge Over the River Kartoa at Ch 4500 on Chatra GC-Gilabari Ghat Via Nischintobati Primary School Road, Road ID: 185764034			District: Rangpur	Upazila: Pirganj
Span/Girder Ref: Span-1, Girder-3	Girder Length (m): 43m	Cable Ref. to be Stressed: 02 of 04		
Girder Casting Date: 22-06-2026	Stressing Stage of Cable : 02		Jacking end Ref:	End 1:
Date of Tensioning: 09-07-2026			End 2:	

Design information:

Dia of Strand (mm): 15.24mm	No. of Strand Cable: 19 nos	Anchorage Brand: Decomate	UTS of Strand (N/mm2): 1860	M. of Elasticity, E (MPa): 197000
Area of Strand (mm2): 140	Area of Cable, A (mm2): 2660	Design Jacking Force P: 3750 Kn		
Design Elongation Each end (mm):	Design Elongation for Gripping Length (mm): N/A	Actual Gripping Length (mm):	Corrected Elongation for Grip Length d (mm):	
Design Cable Slip (mm): 6mm	Design Conc. St. during Tensioning (N/mm2): 36 MPa		Actual Conc. St at the time of Tensioning (N/mm2): 38.06 MPa	

Stressing & Jack information:

Pump Model No. P1: P2:	Pressure gauge Model. M1: M2:	Tensioning Ram Area (m²): TA1: TA2:	Blocking Ram Area (cm²): BA1: BA2:
Actual Area of Cable, A1 (mm²): 2707.50	Modulus of Elasticity, E1 (MPa): 200400	Corrected Elongation (mm) for Actual A1 & E1: $\delta = (A \cdot E / A1 \cdot E1): 145.18$ mm	
Jack Pres. With Jack loss (BAR) Kg/cm²: J1: P/(TA1*efficiency): J2: P/(TA2*efficiency):	Calibrated Jack press (Kg/cm²) CJ1: CJ2:	Initial Jack Press (Kg/cm²) ICJ1: ICJ2:	Initial Marking: IRJ1: IRJ2:
Actual calculated Elongation for Grip Length, d1 (mm): 4.14mm	Gross Slip of Cable (mm): (Final Elong. - Net Elong.)	Net Slip at Jack end after lock-off (mm): (Gross Slip - Calcul. Elong. d1 for Grip length)	
Blocking Pressure Kg/cm²: E1: Auto Block E2: Auto Block	At end 1	At end 2	At end 1
	149-142=7	153-146=7	7-4.14=2.86

Record of Stressing & Elongation:

Avg. % of Design Load	Actual Applied Pressure		Calculated Gauge Pressure Kg/cm2		Reading for Elongation (mm)		Measured Elongation At both Jacking end (mm)			Corre ction Factor For ICJ	Final/ Total Elongation. (mm)	Averag e Elongat ion At each End.	Remarks (Average Slip at each end)
	Col(1)		Col(2)		Col(3)		Col(4)=(Col 3-IRJ)				Col(5)		
	1	2			1	2	1	2	Avg.		1	2	
	KN	TON	Kg/cm2	Kg/cm2	-	-	-	-	-		-	-	
20	750	76.48	87.22	85.87	30	35	-	-	-				2.86+2.86
40	1500	152.96	175.43	175.58	55	60	25	25	25				2
60	2250	229.44	263.65	265.19	75	80	45	45	45				= 2.86
80	3000	305.91	351.86	354.85	102	103	72	68	70				< 6
95	3562.5	363.27	418.01	422.09	129	133	99	98	98.5				OK
98	3675	374.75	431.25	435.09	137	141	107	106	106.5				
100	3750	382.39	440.07	444.50	143	148	113	113	113	23.5			
102	3825	390.04			149	153	119	118	118.5			142	
105	3937.5	401.51											
Lock-off					142	146							

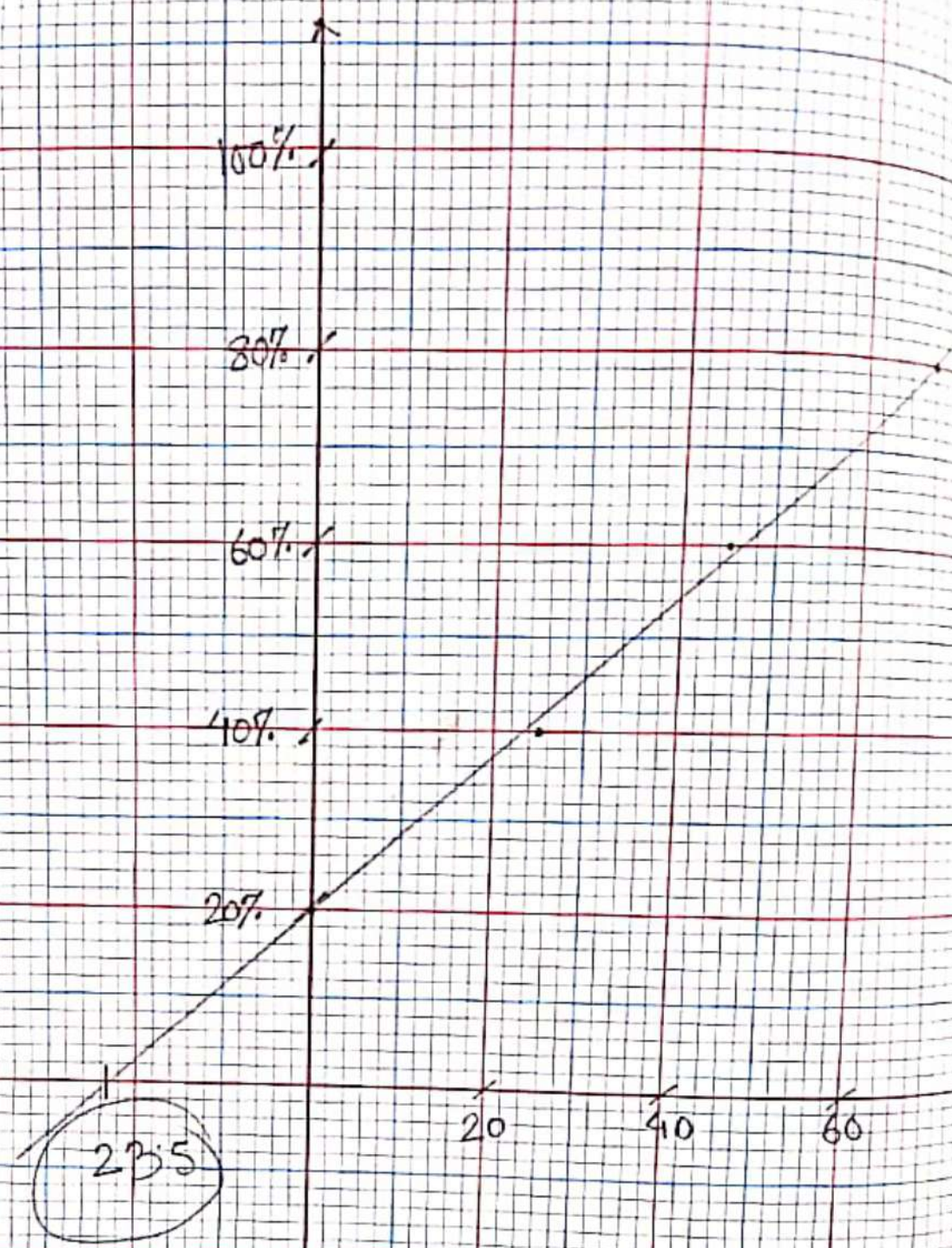
Client's Representative

Consultant's Representative

Contractor's Representative



span



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Span/Girder Ref: Span-1, Girder-1	Girder Length (m): 43m	Cable Ref. to be Stressed 03 of 04		
Girder Casting Date: 22-06-2026	Stressing Stage of Cable : 02		Jacking end Ref:	End 1:
Date of Tensioning: 09-07-2026				End 2:

Design information: 06.08.2026

Dia of Strand (mm): 15.24mm	No. of Strand Cable: 19 nos	Anchorage Brand: Decorate	UTS of Strand (N/mm2): 1860	M. of Elasticity, E (MPa): 197000
Area of Strand (mm2): 140	Area of Cable, A (mm2): 2660	Design Jacking Force P: 3750 Kn		
Design Elongation Each end (mm):	Design Elongation for Gripping Length (mm): N/A	Actual Gripping Length (mm):	Corrected Elongation for Grip Length δ (mm):	
Design Cable Slip (mm): 6mm	Design Conc. St. during Tensioning (N/mm2):36 MPa		Actual Conc. St at the time of Tensioning (N/mm2): 38.06 MPa	

Stressing & Jack information:

Pump Model No. P1: P2:	Pressure gauge Model: M1: M2:	Tensioning Ram Area (m ²): TA1: TA2:	Blocking Ram Area (cm ²): BA1: BA2:
Actual Area of Cable, A1 (mm ²): 2707.50	Modulus of Elasticity, E1(MPa): 200400	Corrected Elongation (mm) for Actual A1 & E1 : $\delta \cdot (A \cdot E / A1 \cdot E1) : 149.04$ mm	
Jack Pres. With Jack loss (BAR) Kg/cm ² . J1: P/(TA1*efficiency): J2: P/(TA2*efficiency):	Calibrated Jack press. (Kg/cm ²) CJ1: CJ2:	Initial Jack Press. (Kg/cm ²) ICJ1: 38 ICJ2: 39	Initial Marking: IRJ1: 38 IRJ2: 39
Actual calculated Elongation for Grip Length, $\delta 1$ (mm): mm	Gross Slip of Cable (mm): (Final Elong. - Net Elong.)	Net Slip at Jack end after lock-off (mm): (Gross Slip - Calcul. Elong. $\delta 1$ for Grip length)	
Blocking Pressure Kg/cm ² : E1: Auto Block E2: Auto Block	At end 1 At end 2	At end 1 At end 2	At end 1 At end 2
	151-144=7 156-148=8	7-4.14=2.86 8-4.14=3.86	

Record of Stressing & Elongation:

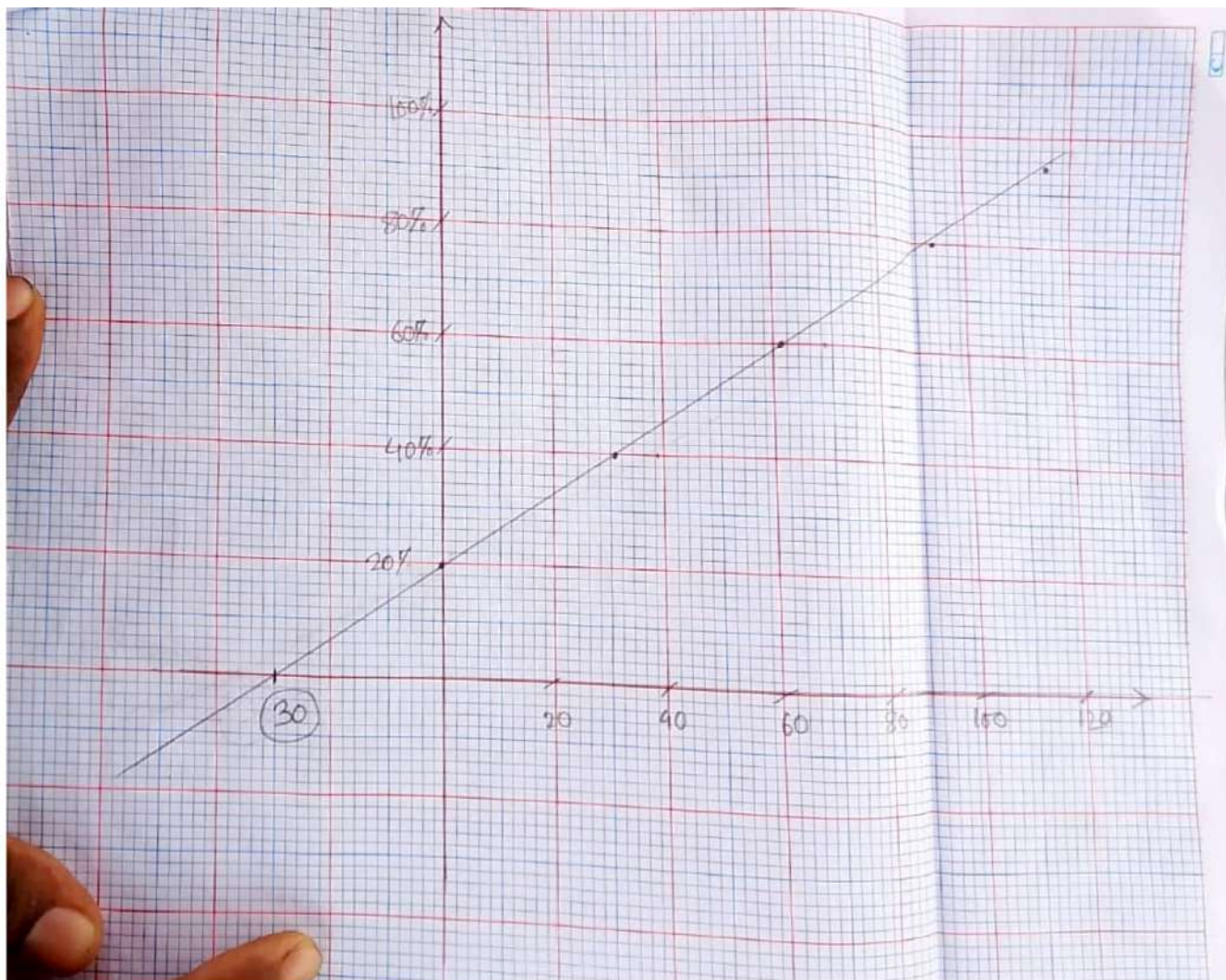
Avg. % of Design Load	Actual Applied Pressure		Calculated Gauge Pressure Kg/cm ²		Reading for Elongation (mm)		Measured Elongation At both Jacking end (mm)			Corre ction Facto r For ICJ	Final/ Total Elong ation. (mm)	Averag e Elongat ion At each End.	Remarks (Average Slip at each end)
	Col(1)		Col(2)		Col(3)		Col(4)=(Col 3-IRJ)			Col (5)	Col(6)= (4+5)		
	1	2			1	2	1	2	Avg.		1	2	
	KN	TON	Kg/cm ²	Kg/cm ²	-	-	-	-	-		-	-	
20	750	76.48	87.22	85.37	38	39	-	-	-				2.86+3.86
40	1500	152.96	175.43	175.53	68	71	30	32	31				2
60	2250	229.44	263.65	265.19	98	101	60	62	61				=3.36
80	3000	305.91	351.86	354.81	129	134	91	95	93				
95	3562.5	363.27	418.01	422.09	151	156	113	117	115	30		145	<6
98	3675	374.75	431.25	43.09									
100	3750	382.39	440.67	444.58									OK
102	3825	390.04											
105	3937.5	401.51											
Lock -off					144	148							

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Span/Girder Ref: Span-1, Girder-A	Girder Length (m): 43m		Cable Ref. to be Stressed: 04 of 04	
Girder Casting Date: 18-07-2026	Stressing Stage of Cable : 02		Jacking end Ref: L 709047	End 1: L 709047
Date of Tensioning: 08-08-2026			End 2: L 709046	

Design information:

Dia of Strand (mm): 15.24mm	No. of Strand Cable: 19 nos	Anchorage Brand: Decomate	UTS of Strand (N/mm2): 1860	M. of Elasticity, E (MPa): 197000
Area of Strand (mm2): 140	Area of Cable, A (mm2): 2660	Design Jacking Force P: 3750 Kn		
Design Elongation Each end (mm):	Design Elongation for Gripping Length (mm): N/A	Actual Gripping Length (mm):	Corrected Elongation for Grip Length δ (mm):	
Design Cable Slip (mm): 6mm	Design Conc. St. during Tensioning (N/mm2): 36 MPa		Actual Conc. St at the time of Tensioning (N/mm2): 38.06 MPa	

Stressing & Jack information:

Pump Model No. P1: 710567 P2: 710570	Pressure gauge Model: M1: M2:	Tensioning Ram Area (m2): TA1: TA2:	Blocking Ram Area (Cm2): BA1: BA2:
Actual Area of Cable, A1 (mm2): 2707.50	Modulus of Elasticity, E1 (MPa): 200400	Corrected Elongation (mm) for Actual A1 & E1: $\delta \cdot (A \cdot E / A1 \cdot E1)$: 149.04 mm	
Jack Pres. With Jack loss (BAR) Kg/cm2. J1: P/(TA1 * efficiency): J2: P/(TA2 * efficiency):	Calibrated Jack press. (Kg/cm2) CJ1: CJ2:	Initial Jack Press. (Kg/cm2) ICJ1: 87.22 ICJ2: 85.87	Initial Marking: IRJ1: 36 IRJ2: 41
Actual calculated Elongation for Grip Length, δ 1 (mm): 4.14 mm	Gross Slip of Cable (mm): (Final Elong. - Net Elong.)		Net Slip at Jack end after lock-off (mm): (Gross Slip - Calcul. Elong. δ 1 for Grip length)
Blocking Pressure Kg/Cm2: E1: Auto Block E2: Auto Block	At end 1	At end 2	At end 1 At end 2
	153-147=6	152-145=7	6-4.14=1.86 7-4.14=2.86

Record of Stressing & Elongation:

Record of Stressing & Elongation:															
Avg. % of Design Load	Actual Applied Pressure		Calculated Gauge Pressure Kg/cm2		Reading for Elongation (mm)		Measured Elongation At both Jacking end (mm)			Corre ction Facto r For ICJ	Final/ Total Elongation. (mm)	Averag e Elongat ion At each End.	Remarks (Average Slip at each end)		
	Col(1)		Col(2)		Col(3)		Col(4)=(Col 3-IRJ)							Col (5)	Col(6)= (4+5)
	1	2			1	2	1	2	Avg.						
	KN	TON	Kg/cm2	Kg/cm2	-	-	-	-	-		-	-			
20	750	76.48	87.22	85.87	36	41	-	-	-					1.86+2.86	
40	1500	152.96	175.43	175.53	65	70	29	29	29					2	
60	2250	229.44	263.65	265.19	95	99	59	58	58.5					=2.36	
80	3000	305.91	351.86	354.85	125	124	89	83	86						
95	3562.5	363.27	418.01	422.09	158	152	117	111	114	30			144	2.6	
98	3675	374.75	431.25	435.89											
100	3750	382.39	440.07	444.50											
102	3825	390.04	448.89	453.47											
105	3937.5	401.51	462.12	466.92											
Lock -off					147									OK	

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